EMOTIA • Walkthrough & Architecture

Emotion-Aware Intelligent Desk Assistant for Human-Robot Interaction Research
Paper: *Context-Aware and Explainable Emotion Intelligence for a Personalized HRI Desk Assistant*

Live Static Site: https://a-sm20.github.io/EMOTIA/	**GitHub Repository:** https://github.com/A-SM20/EMOTIA
Tech Stack: React 19, Tailwind CSS, Recharts, Lucide Icons	**Local Dev Server:** http://localhost:5173/

1. Executive Overview & Visual Identity

EMOTIA is an intelligent workstation assistant frontend developed to evaluate context-aware affective computing in desk-bound Human-Robot Interaction. It demonstrates how real-time facial feature tracking, speech prosody extraction, and developmental desktop context fuse via cross-attention into an explainable, proactive robotic desk companion.
Visual Identity: Editorial, ultra-modern monochrome aesthetic (deep #09090b matte dark surface, clean #121215 card surfaces, hairline border-white/[0.08] boundaries, and **Plus Jakarta Sans** typography).

2. Application Screens & Key Features

Screen 1: Dashboard (Home / Live Assistant)

- **Dynamic Facial Perception HUD:** Real-time animated 68-point facial landmark wireframe with organic mood-driven movement (AU4 brow furrowing, relaxed breathing sways, smiling contours, and ocular saccades).
- **Current Affect State Card:** Live emotion label, confidence meter, Russell's 2D Circumplex Model telemetry (Valence & Arousal), and active task context (VS Code / PyTorch).
- **Proactive AI Assistant Card:** Contextual guidance bubble, Web Speech API (TTS) voice synthesis, and interactive prompt input bar with ambient microphone sensing.
- **System Status Matrix:** Real-time health status and latencies for 9 core submodules (Vision, Acoustic, Fusion, Context, Memory, LLM, Backend).

Screen 2: Live Emotion (Multimodal Fusion Monitor)

- **3 Parallel Signal Streams:** Side-by-side readouts for Facial Vision (Swin-FER), Acoustic Speech (Wav2Vec2), and Fused Prediction (Cross-Modal Attention Matrix) highlighting the multimodal fusion contribution.
- **Emotion Probability Spectrum & Rolling History:** Horizontal animated distribution bars across 7 discrete affective states and a live Recharts visualizer tracking historical intensity fluctuations.

Screen 3: Memory (Personalized Profile & Habits)

- **Researcher Persona Card:** Dr. Alex Vance (Senior HRI Researcher), preferred interaction style, and domain ontology tags.
- **Longitudinal Trajectory & Learned Habits:** Hourly daily chart showing Calm vs. Focus vs. Stress distributions alongside prioritized behavioral rules with explicit research disclaimer.

Screen 4: Conversations (Transcript History)

- **Chronological Dialog Stream:** 8 rich exchanges tagged with metadata chips (*'Detected: Frustrated • 87%'* and *'Context: Programming'*) with interactive prompt testing.

Screen 5: Insights (Transparent Explainability / XAI)

- **Why Did the System Predict This?:** Feature attribution breakdown quantifying percentage weights (Facial AU, Acoustic Prosody, Context, Trajectory) alongside dynamic natural-language explanations.
- **System Architecture Strip:** 5-stage horizontal pipeline (User & Context → Multimodal Perception → Context & Memory → Decision/XAI → Assistant UI) tagged with module owner teams.

3. Faculty & Evaluator Preset Scenarios

A 1-click segmented pill bar at the top of the interface synchronizes the entire application across 5 distinct HRI affective scenarios:

Scenario	Affect & Conf.	Desk Context	Proactive Assistant Action
■ **Frustrated**	Frustrated (87%) Val: -0.62 | Aro: +0.74	VS Code • 3.2 hrs active Repeated syntax/tensor error	Offers step-by-step problem breakdown and automated tensor shape inspection.
■ **Calm**	Calm / Flow (94%) Val: +0.72 | Aro: -0.35	Overleaf • LaTeX Section 4: Attention Fusion	Maintains silent background monitoring; suppresses non-urgent notification alerts.

■ **Happy**	Happy (91%) Val: +0.88 | Aro: +0.65	JupyterLab • Benchmark Ablation score: 94.8% F1	Celebrates milestone; offers automated ROC curve and LaTeX table export.
■ **Stressed**	Stressed (89%) Val: -0.78 | Aro: +0.89	CMT Conference Portal Deadline in 2 hours	Streamlines secondary tasks; auto-checks PDF formatting and bibliography DOIs.
■ **Neutral**	Neutral (84%) Val: +0.05 | Aro: +0.10	Acrobat Reader ArXiv HRI Research PDF	Offers smart paper summarization and citation reference extraction.

4. System Architecture & CI/CD Deployment

Deployment Pipeline: Deployed as a static SPA on GitHub Pages using an automated GitHub Actions workflow (.github/workflows/deploy.yml). Pushes to main trigger Vite production builds and publish updates in ~35 seconds.
State Architecture: Centralized React Context (EmotionContext) synchronizing scenario state, speech synthesis engines, real-time animation loops, and rolling Recharts telemetry across all views.